

Credit Card Default Auto-ML Report

🕒 2026-05-28 21:55:38 UTC

🏆 Best: **catboost** roc_auc = 0.7796

📊 학습 24000행

테스트 6000행

Feature 197개

1 모델 비교 (holdout 기준)

model	roc_auc	ks	lift	accuracy	precision	recall	f1	pr_auc	best_iter (avg)
lgbm	0.7746	0.4248	3.1123	0.8137	0.7061	0.2698	0.3904	0.5430	154
xgb	0.7777	0.4265	3.1123	0.8195	0.6723	0.3587	0.4678	0.5548	175
catboost BEST	0.7796	0.4362	3.1274	0.8177	0.6589	0.3640	0.4689	0.5588	305
elasticnet	0.7314	0.3895	2.9314	0.8072	0.6580	0.2668	0.3796	0.4988	-
ensemble	0.7663	0.4168	3.0595	0.8153	0.6585	0.3429	0.4509	0.5377	-

2 CV (OOF) 비교

model	roc_auc	ks	lift	accuracy	precision	recall	f1	pr_auc
lgbm	0.7848	0.4359	3.1795	0.8212	0.6859	0.3537	0.4668	0.5568
xgb	0.7851	0.4325	3.2002	0.8217	0.6843	0.3598	0.4716	0.5560
catboost BEST	0.7841	0.4356	3.1946	0.8220	0.6769	0.3733	0.4812	0.5586
elasticnet	0.7463	0.4038	2.9591	0.8084	0.6687	0.2654	0.3800	0.5075
ensemble	0.7753	0.4245	3.1155	0.8190	0.6804	0.3428	0.4559	0.5465

3 오버핏 점검 (Train vs Holdout)

metric	lgbm			xgb			catboost BEST				
	Train	Test	Δ	Train	Test	Δ	Train	Test	Δ	Train	Test
roc_auc	0.8273	0.7746	+0.0527	0.8222	0.7777	+0.0445	0.8099	0.7796	+0.0303	0.7482	0.7314
ks	0.5022	0.4248	+0.0775	0.4841	0.4265	+0.0576	0.4660	0.4362	+0.0297	0.4062	0.3895
lift	3.2530	3.1123	+0.1407	3.3189	3.1123	+0.2066	3.4037	3.1274	+0.2763	2.9780	2.9314
accuracy	0.8188	0.8137	+0.0052	0.8278	0.8195	+0.0083	0.8308	0.8177	+0.0131	0.8086	0.8072
precision	0.7399	0.7061	+0.0338	0.7078	0.6723	+0.0355	0.7133	0.6589	+0.0543	0.6702	0.6580
recall	0.2791	0.2698	+0.0094	0.3773	0.3587	+0.0186	0.3931	0.3640	+0.0291	0.2652	0.2668

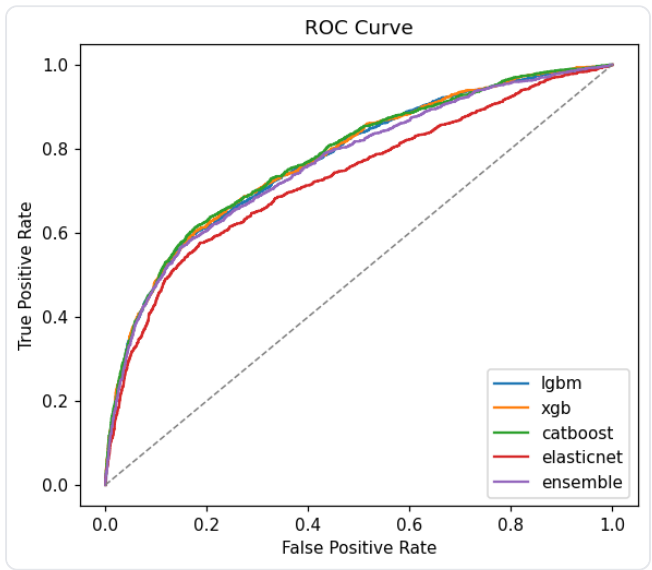
metric	lgbm			xgb			catboost			BEST	
	Train	Test	Δ	Train	Test	Δ	Train	Test	Δ	Train	Test
f1	0.4054	0.3904	+0.0150	0.4922	0.4678	+0.0244	0.5069	0.4689	+0.0379	0.3800	0.379
pr_auc	0.5999	0.5430	+0.0569	0.6174	0.5548	+0.0626	0.6194	0.5588	+0.0606	0.5102	0.498

Δ = Train - Test. Train 은 final 모델을 학습 데이터에 fit 한 뒤 동일 학습 데이터로 in-sample 스코어링한 결과로, 과도적으로 over-optimistic 한 추정치다. Test 와의 양의 gap 이 클수록 메모리제이션(overfit) 신호이며, $|\Delta| \geq 0.05$ / $|\Delta| \geq 0.10$ 인 셀은 색으로 강조한다.
 (정확한 임계치는 도메인/지표에 따라 다르며 본 색은 참고용.)

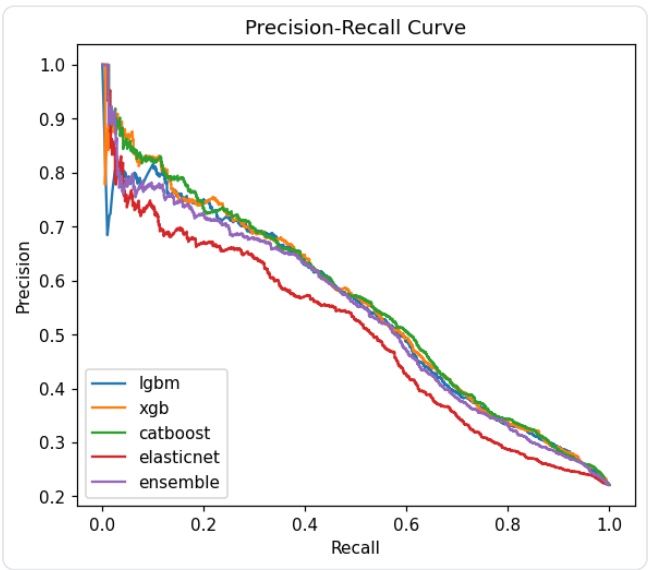
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곡선 비교

ROC Curve



Precision-Recall Curve



5 Best 모델 상세 — catboost

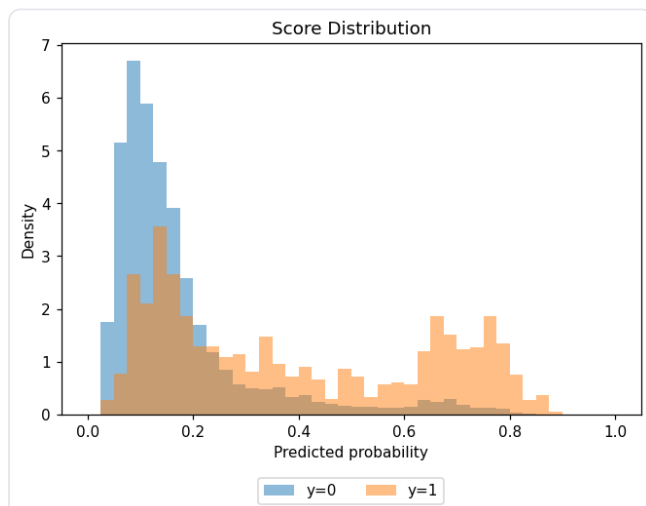
Feature Importance (relative, top 19 · 전체 분포는 feature_importance.csv)

Rank	Feature	Importance
1	PAY_0	0.2307
2	LIMIT_BAL	0.1047
3	BILL_AMT1	0.0909
4	PAY_2	0.0574
5	PAY_3	0.0499
6	PAY_AMT3	0.0471
7	PAY_AMT2	0.0468
8	BILL_AMT2	0.0409
9	PAY_AMT1	0.0408
10	PAY_4	0.0395
11	AGE	0.0342
12	PAY_AMT6	0.0327
13	PAY_6	0.0313
14	PAY_5	0.0298
15	PAY_AMT4	0.0285

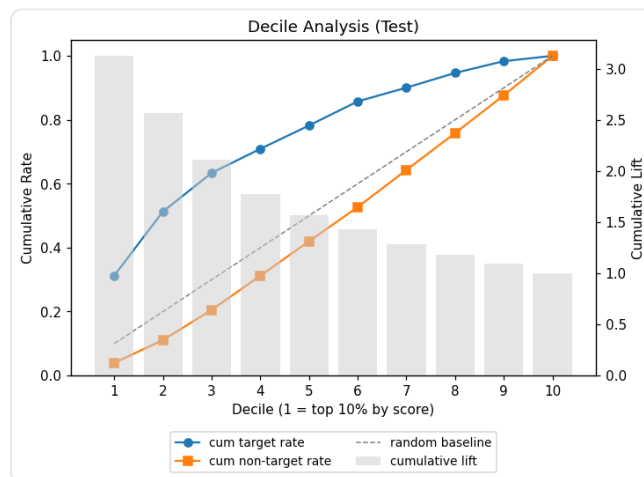
Rank	Feature	Importance
16	BILL_AMT5	0.0259
17	BILL_AMT6	0.0255
18	PAY_AMT5	0.0220
19	SEX	0.0213

10-분위 분석 (테스트셋, 점수 내림차순) 및 점수 분포

Score Distribution by Label



Decile Analysis Chart



Decile	Score Range	Count	Target	Non-Target	Cum Target	Cum NonTarget	Cum Count	Cum %
1	0.596 ~ 0.886	600	415	185	415	185	600	10.0%
2	0.324 ~ 0.596	600	266	334	681	519	1200	20.0%
3	0.213 ~ 0.324	600	160	440	841	959	1800	30.0%
4	0.170 ~ 0.213	600	100	500	941	1459	2400	40.0%
5	0.144 ~ 0.169	600	97	503	1038	1962	3000	50.0%
6	0.122 ~ 0.144	600	100	500	1138	2462	3600	60.0%
7	0.103 ~ 0.122	600	57	543	1195	3005	4200	70.0%
8	0.086 ~ 0.103	600	61	539	1256	3544	4800	80.0%
9	0.067 ~ 0.086	600	49	551	1305	4095	5400	90.0%
10	0.029 ~ 0.067	600	22	578	1327	4673	6000	100.0%

Confusion Matrix threshold = 0.5

	Pred 0	Pred 1
True 0	4423	250
True 1	844	483

6 학습 설정 요약

설정 키	값
target_column	default
categorical_columns	SEX, EDUCATION, MARRIAGE
id_columns	client_id
preprocessing.numeric_null_strategy	median
preprocessing.categorical_null_strategy	most_frequent
preprocessing.outlier_method	percentile
preprocessing.outlier_action	clip
preprocessing.skew_method	signed_log1p
preprocessing.skew_threshold	1.0
preprocessing.scaling_method	standard
training.cv_folds	4
training.early_stopping_rounds	50
training.primary_metric	roc_auc

설정 키	값
training.random_state	42
tuning.enabled	True
tuning.n_trials	15
tuning.cv_folds	3
tuning.timeout	None

7 모델 파라미터

lgbm 튜닝 완료 · 15회 best OOF 0.7849	
파라미터	값
objective	binary
metric	auc
verbose	-1
n_estimators	800
learning_rate	0.021471858797027
num_leaves	85
min_data_in_leaf	55
feature_fraction	0.7779717392187493
bagging_fraction	0.6052561262871233
reg_alpha	0.07115953064092197
reg_lambda	7.860792250906604
min_gain_to_split	0.1471710061864095

xgb 튜닝 완료 · 15회 best OOF 0.7855	
파라미터	값
objective	binary:logistic
eval_metric	auc
tree_method	hist
n_estimators	800
learning_rate	0.026000059117302653
max_depth	7
subsample	0.6563696899899051

파라미터	값
colsample_bytree	0.9208787923016158
min_child_weight	1.6709557931179373
reg_alpha	7.620481786158549
reg_lambda	0.08916674715636537
gamma	0.993578407670862

catboost BEST 튜닝 완료 · 15회 best OOF 0.7847	
파라미터	값
loss_function	Logloss
eval_metric	AUC
iterations	800
verbose	False
allow_writing_files	False
learning_rate	0.03220579752389847
depth	6
l2_leaf_reg	4.60313968685407
random_strength	0.723236835259315
bagging_temperature	0.6449759616213921

elasticnet 튜닝 완료 · 15회 best OOF 0.7461	
파라미터	값
max_iter	1000
C	7.553503645583172
l1_ratio	0.005297120813991926

ensemble 튜닝 미수행	
파라미터	값
weights	{'elasticnet': 0.4840579071352653, 'catboost': 0.5159420928647347}

8 변수 선택 (Stability Selection)

base: lasso

부분표본 80회

임계값 0.60

채택 19 / 23

표 노출 23 / 전체 23 (전체 분포는 feature_selection.csv)

Feature	Selection Frequency		채택
LIMIT_BAL		100.0%	✓ 채택
AGE		100.0%	✓ 채택
PAY_0		100.0%	✓ 채택
PAY_2		100.0%	✓ 채택
PAY_3		100.0%	✓ 채택
PAY_4		100.0%	✓ 채택
BILL_AMT1		100.0%	✓ 채택
PAY_AMT1		100.0%	✓ 채택
PAY_AMT2		100.0%	✓ 채택
PAY_AMT3		100.0%	✓ 채택
PAY_AMT6		100.0%	✓ 채택
PAY_5		98.8%	✓ 채택
BILL_AMT5		98.8%	✓ 채택
PAY_6		97.5%	✓ 채택
PAY_AMT4		96.2%	✓ 채택
PAY_AMT5		95.0%	✓ 채택
SEX		91.2%	✓ 채택
BILL_AMT6		88.8%	✓ 채택
BILL_AMT2		71.2%	✓ 채택
BILL_AMT3		56.2%	제외
BILL_AMT4		28.7%	제외
EDUCATION		21.2%	제외
MARRIAGE		15.0%	제외